Chapter 1 : Sequences and Series

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I- Introduction to Sequences

Investigation

Around 1260 AD, the Kurdish historian Ibn Khallikān recorded the following story about Sissa ibn Dahir and a chess game against the Indian King Shihiram. (The story is also told in the Legend of the Ambalapuzha Paal Payasam, where the Lord Krishna takes the place of Sissa ibn Dahir, and they play a game of chess with the prize of rice grains rather than wheat.)

> King Shihiram was a tyrant king, and his subject Sissa ibn Dahir wanted to teach him how important all of his people were. He invented the game of chess for the king, and the king was greatly impressed. He insisted on Sissa ibn Dahir naming his reward, and the wise man asked for one grain of wheat for the first square, two grains of wheat for the second square, four grains of wheat for the third square, and so on, doubling the wheat on each successive square on the board.
>
> The king laughed at first and agreed, for there was so little grain on the first few squares. By halfway he was surprised at the amount of grain being paid, and soon he realised his great error : that he owed more grain than there was in the world.

Things to think about :

1. How can we describe the number of grains of wheat for each square ?
2. What expression gives the number of grains of wheat for the nth square ?
3. Find the total number of grains of wheat that the king owed.

Definition

A number sequence is a pattern of numbers arranged in a particular order according to a rule.

Example

Here are some number sequences :

$$8, 11, 14, 17, \ldots \qquad 800, 400, 200, 100, \ldots \qquad 1, 4, 9, 16, 25, \ldots \qquad 5, 10, 15, 20, 25, \ldots$$

Definition

Each individual number, or element, of a sequence is called a **term**.

You can also use the notation u_n to denote the nth term of a sequence, where n is a positive integer.

Example

So for $8, 11, 14, 17, \ldots$ you could say

$$u_1 = 8, \quad u_2 = 11, \quad u_3 = 14, \quad \ldots$$

For this sequence, you could write

$$u_1 = 8, \quad u_{n+1} = u_n + 3$$

Definition

This is called a **recursive formula**, in which the value of a term depends on the value of the previous term.

Definition

A sequence which continues forever is called an **infinite sequence**. A sequence which terminates is called a **finite sequence**.

The general term u_n is a function where $n \mapsto u_n$. The domain is $n \in \mathbb{Z}^+$ for an infinite sequence or a subset of $\mathbb{Z}^+$ for a finite sequence.

Example

$$\{2n + 1\} \quad \Rightarrow \quad 3, 5, 7, 9, 11, \ldots$$

II - Arithmetic Sequences

Definition

In an arithmetic sequence, the terms increase or decrease by a constant value. This value is called the **common difference**, or d. The common difference can be positive or negative.

Example

$$8, 11, 14, 17, \ldots \quad u_1 = 8, \ d = 3$$

$$35, 30, 25, 20, \ldots \quad u_1 = 35, \ d = -5$$

$$4, 4.1, 4.2, 4.3, \ldots \quad u_1 = 4, \ d = 0.1$$

Property

Recursive formula of any arithmetic sequence

$$u_{n+1} = u_n + d$$

Property

General formula of any arithmetic sequence

You can find the nth term using :

$$u_n = u_1 + (n-1)d$$

Example

1. An arithmetic sequence has first term 19 and 15th term 31.6. Find the common difference.
2. In an arithmetic sequence, $u_{10} = 37$ and $u_{21} = 4$. Find the common difference and the first term.
3. Find the value of x in the arithmetic sequence $3, x, 8, \ldots$
4. Find the value of m in the arithmetic sequence $m, 13, 3m-6, \ldots$

III- Geometric sequences

a.Formulas

Definition

In a geometric sequence, each term can be obtained by multiplying the previous term by a constant value. This value is called the **common ratio**, or r.

Remark. The common ratio r can be positive or negative.

Example

$$1, 5, 25, 125, \ldots \quad u_1 = 1, \ r = 5$$

$$3, -6, 12, -24, \ldots \quad u_1 = 3, \ r = -2$$

Property

Recursive formula of any geometric sequence

$$u_{n+1} = u_n \cdot r$$

You can find any term of the sequence by multiplying the previous term by the common ratio r.

Property

General formula of any geometric sequence

$$u_n = u_1 \, r^{n-1}$$

Example

At 8 :00 a.m., 1000 mg of medicine is given to a patient. At the end of each hour, the amount of medicine in the patient's bloodstream is 60% of the amount present at the beginning of the hour.

(a) What portion of the medicine remains in the patient's bloodstream at 12 noon if no additional medication had been given ?

(b) If a second dose of 1000 mg is given at 10 :00 a.m., what is the total amount of the medication in the patient's bloodstream at 12 noon ?

b.Compound interest

Compound interest is an example of a geometric sequence.

When we borrow money we pay interest, and when we invest money we receive interest. Suppose an amount of €1000 is put into a savings account that has an annual interest rate of 6%. How much money will we have in the bank at the end of 4 years ?

It is important to note that the 6% interest is given annually and is added to the savings account, so that in the following year it will also earn interest, and so on.

Time in years	Amount in the account (€)
0	1000
1	$1000 + 1000 \times 0.06 = 1000(1 + 0.06)$
2	$1000(1 + 0.06) + (1000(1 + 0.06)) \times 0.06 = 1000(1 + 0.06)^2$
3	$1000(1 + 0.06)^2 + (1000(1 + 0.06)^2)(0.06) = 1000(1 + 0.06)^3$
4	$1000(1 + 0.06)^3(1 + 0.06) = 1000(1 + 0.06)^4$

Property

General Compound Interest Formula Let

$$A_0 = P \quad \text{initial amount,} \quad r = \text{annual interest rate,} \quad t = \text{number of years.}$$

Then the amount after t years is

$$A_t = P(1+r)^t$$

Proof

Time in years	Amount in the account
0	$A_0 = P$
1	$A_1 = P(1+r)$
2	$A_2 = P(1+r)^2$
$\vdots$	$\vdots$
t	$A_t = P(1+r)^t$

Property

Compound Interest with Multiple Compounding Periods Interest compounded n times per year :

$$A = P\left(1+\frac{r}{n}\right)^{nt}$$

Example

€1000 is invested in an account paying 6% compound interest per annum. Find the amount after 10 years if compounding is :

1. annual
2. quarterly
3. monthly

Example

1. Victoria put €1500 into a savings account at 4% annual interest compounded semiannually. Find the value after 10 years.
2. Charlotte deposited £500 at 4% compounded quarterly at the birth of her daughter. How much will she have on the daughter's 16th birthday ?
3. How much must be invested now to have €4000 after 6 years if interest is 5% compounded quarterly ?
4. A town's population in 2017 was 7554. If it grows at 0.5% annually, find the population in 2022.
5. You invest €1000. What annual rate is needed to double it in 10 years if interest is compounded quarterly ?

IV- Series

a. Sigma notation

This section looks at ways to add the terms of a sequence. Adding the terms of a sequence gives a **series**.

$$u_1, u_2, u_3, \ldots, u_n \quad \text{is a sequence.}$$

$$u_1 + u_2 + u_3 + \cdots + u_n \quad \text{is a series.}$$

Definition

The Greek letter Σ, called "sigma", is often used to represent sums of values.

$$u_1 + u_2 + u_3 + \cdots + u_n = \sum_{k=1}^{n} u_k$$

$\sum_{k=1}^{n} u_k$ reads "the sum of all numbers of the form u_k where $k = 1,2,3,\ldots,n$".

Example

Consider the sequence $1, 4, 9, 16, 25, \ldots$

1. Write down an expression for S_n.
2. Find S_n for $n = 1,2,3,4,5$.

Property

$$\sum_{k=1}^{n}(a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$$

$$\sum_{k=1}^{n} ca_k = c\sum_{k=1}^{n} a_k \quad \text{if } c \text{ is a constant.}$$

$$\sum_{k=1}^{n} c = cn$$

b. Arithmetic series

Definition

An arithmetic series is the sum of the terms of an arithmetic sequence.

Property

The sum of a finite arithmetic series with first term u_1, common difference d, and last term u_n, is :

$$S_n = \frac{n}{2}(u_1 + u_n) \quad \text{or} \quad S_n = \frac{n}{2}(2u_1 + (n-1)d)$$

Proof

If the first term is u_1 and the common difference is d, the terms are

$$u_1,\ u_1+d,\ u_1+2d,\ u_1+3d,\ldots$$

Suppose that u_n is the final term of an arithmetic series.

So,

$$S_n = u_1 + (u_1+d) + (u_1+2d) + \cdots + (u_n-2d) + (u_n-d) + u_n$$

But

$$S_n = u_n + (u_n-d) + (u_n-2d) + \cdots + (u_1+2d) + (u_1+d) + u_1$$

(reversing them).

Adding these two equations vertically we get

$$2S_n = (u_1+u_n) + (u_1+u_n) + \cdots + (u_1+u_n)$$

(n of these).

Therefore,

$$2S_n = n(u_1+u_n)$$

$$\therefore\ S_n = \frac{n}{2}(u_1+u_n) \quad \text{where} \quad u_n = u_1+(n-1)d$$

Thus, the sum of a finite arithmetic series with first term u_1, common difference d, and last term u_n, is :

$$S_n = \frac{n}{2}(u_1+u_n) \quad \text{or} \quad S_n = \frac{n}{2}(2u_1+(n-1)d)$$

Property

Sum of the First n Integers

$$1+2+3+\cdots+n = \frac{n(n+1)}{2}, \qquad n \in \mathbb{Z}^+.$$

Example

Find the sum of

$$4+7+10+13+\cdots$$

to 50 terms.

c. Geometric Series

Definition

A geometric series is the sum of the terms of a geometric sequence.

If the first term is u_1 and the common ratio is r, then the terms are :

$$u_1,\ u_1 r,\ u_1 r^2,\ u_1 r^3, \ldots$$

So,

$$S_n = u_1 + u_1 r + u_1 r^2 + u_1 r^3 + \cdots + u_1 r^{n-2} + u_1 r^{n-1}.$$

Property

For a finite geometric series with $r \neq 1$:

$$S_n = \frac{u_1(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{u_1(1 - r^n)}{1 - r}.$$

Proof

If

$$S_n = u_1 + u_1 r + u_1 r^2 + u_1 r^3 + \cdots + u_1 r^{n-2} + u_1 r^{n-1},$$

then

$$rS_n = (u_1 r + u_1 r^2 + u_1 r^3 + \cdots + u_1 r^{n-1}) + u_1 r^n.$$

So,

$$rS_n - S_n = u_1 r^n - u_1,$$

$$S_n(r - 1) = u_1(r^n - 1),$$

$$S_n = \frac{u_1(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{u_1(1 - r^n)}{1 - r}, \quad \text{provided } r \neq 1.$$

In the case $r = 1$, we have a sequence in which all terms are the same, and

$$S_n = u_1 n.$$

Example

Find the sum of

$$2 + 6 + 18 + 54 + \cdots$$

to 12 terms.

d. Infinite geometric series

Investigation

Here are three geometric series :

1. $2 + 1 + 0.5 + \cdots$
2. $75 + 30 + 12 + \cdots$
3. $240 - 60 + 15 - 3.75 + \cdots$

1. For each of these series :
   (a) Find the common ratio r.
   (b) Use your GDC to calculate the values of S_{10}, S_{15}, S_{20}. Write the full values you see on your GDC screen.
2. Do you notice any patterns ? Why do you think this is happening ?
3. Now use your GDC to calculate the value of S_{50} for each series. Do you think your calculator is correct ? Explain why or why not.

Paradox Suppose you are walking down a 30-metre hallway. Every ten seconds, you walk half the distance to the end of the hallway. How long will it take you to reach the end of the hallway ? Will you ever get there ?

Example

If $|r| > 1$, the series is said to be divergent and the sum becomes infinitely large.

For example, when $r = 2$,

$$1 + 2 + 4 + 8 + 16 + \cdots$$

is infinitely large.

If $|r| < 1$, or in other words $-1 < r < 1$, then as n becomes very large, r^n approaches 0.

For example, when $r = \frac{1}{2}$,

$$1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \cdots = 2.$$

This means that S_n will get closer and closer to

$$\frac{u_1}{1 - r}.$$

Theorem

If $|r| < 1$, an infinite geometric series of the form

$$u_1 + u_1 r + u_1 r^2 + \cdots = \sum_{k=1}^{\infty} u_1 r^{k-1}$$

will converge to the sum

$$S = \frac{u_1}{1-r}.$$

We call this the *limiting sum* of the series.

Example

Write $0.\overline{7}$ as a rational number

$$0.\overline{7} = \frac{7}{10} + \frac{7}{100} + \frac{7}{1000} + \frac{7}{10000} + \cdots$$

which is a geometric series with infinitely many terms.

In this case, $u_1 = \frac{7}{10}$ and $r = \frac{1}{10}$.

$$S = \frac{u_1}{1-r} = \frac{\frac{7}{10}}{1-\frac{1}{10}} = \frac{7}{9}$$

Therefore,

$$0.\overline{7} = \frac{7}{9}.$$